

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 23

20-07-2026

23. WRITE THE PROLOG PROGRAM TO IMPLEMENT FAMILY TREE.

AIM

To write and execute a Prolog program to implement a family tree and determine relationships such as parent, child, grandparent, and sibling. The program demonstrates how family relationships can be represented using facts and derived using logical rules.

ALGORITHM

1. Start the program.
2. Define parent-child relationships as facts.
3. Define a rule to identify grandparents.
4. Define a rule to identify siblings.
5. Define additional rules for brother and sister relationships.
6. Query the family tree for a particular relationship.
7. Display the result.
8. Stop the program.

PROLOG PROGRAM

% Write a Prolog program to implement Family Tree.

parent(ramesh, rahul).

parent(ramesh, priya).

parent(sita, rahul).

parent(sita, priya).

parent(rahul, arjun).
parent(rahul, ananya).

grandparent(X, Z) :-
 parent(X, Y),
 parent(Y, Z).

sibling(X, Y) :-
 parent(P, X),
 parent(P, Y),
 X $\neq$ Y.

male(ramesh).
male(rahul).
male(arjun).

female(sita).
female(priya).
female(ananya).

brother(X, Y) :-
 sibling(X, Y),
 male(X).

sister(X, Y) :-

sibling(X, Y),

female(X).

SAMPLE INPUT

?- grandparent(ramesh, arjun).

SAMPLE OUTPUT

true.

RESULT

Thus, the family tree was successfully implemented in Prolog. The program correctly identified relationships such as grandparent and sibling by applying logical rules to the family facts.